

The Air-Gapped Workbench: what to change to win

Adversarial technical review | SIH / MRPL team | 6 September 2026

My verdict: keep the inspection-to-approval workflow, but redesign the evidence and decision path. Your best opportunity is reliable, auditable refinery work on modest hardware. Generic local agents, routing and file generation already have substantial competition.

The brief makes good delivery choices: build a complete vertical slice early, reuse an interface, use templates, test local deployment, and include a deliberately blocked network request. Preserve those. The research does not support treating the proposed stack as a settled technical optimum or its six-item interpretation as an independently verified marking scheme.

The most valuable improvement is not a larger reasoning model. It is a **structured evidence record** between OCR and reasoning: each measurement retains its original text, image location, unit, equipment association and applicable threshold. Deterministic code evaluates explicit approved rules. The model explains the result and drafts the note. A human resolves ambiguous critical readings.

Rank	Replace or challenge	Expected advantage to test
1	LLM-only escalation with typed facts and explicit rules	Fewer missed threshold violations; reproducible decisions
2	Whole-page OCR alone with cell-based extraction and selective rereading	Better digit, unit and row association; focused review
3	Graph-plus-vector as the initial retrieval design with lexical + vector search	Better exact clause/tag matching with less indexing work
4	Thinker/Workers/Verifier as the default with a bounded workflow and one agent	Lower latency, easier recovery and fewer uncontrolled loops
5	A pool of resident specialists with staged GPU loading	More usable context and fewer memory failures on 12GB
6	Generic novelty claims with comparison against existing applications	A defensible product advantage judges can inspect

These are engineering hypotheses, not measured wins. The report specifies how to falsify them. **No model, OCR pipeline, security configuration or benchmark was run on your RTX 4070.**

This review covers all eleven sections of the supplied team brief. Only that pasted attachment was available; the referenced project docs, downloaded PDFs, code, packet logs and synthetic corpus were not supplied. Their completion status is therefore team-reported. Official SIH wording could not be independently retrieved. Findings use primary papers, repositories, model cards and product documentation checked on 6 September 2026.

Sources: Mintplex Labs, current documentation, [AnythingLLM Model Router](#) and [Document Generation Agent](#); documented capability establishes competition, not equivalence in reliability.

Correct the claims before defending the architecture

“Nothing deployable exists” is the clearest factual weakness. AnythingLLM already documents automatic model selection using calculated rules or an LLM classifier, including self-hosted deployment. Its built-in document agent produces Word, Excel, PowerPoint and PDF files. Those features overlap directly with the brief's proposed differentiators. Its MIT source license is verified; that does not establish that every optional service or model is covered by MIT.

The Onyx dismissal is incorrect. Official documentation describes sandboxed Python, file inputs, downloadable file outputs, document generation and custom-agent integration. Current release notes also establish image handling. Onyx's Enterprise provider gateway is not proof of an equivalent automatic task classifier; keep that distinction precise. Its FOSS and Enterprise features must not be collapsed into one licensing claim.

The six requirements are useful acceptance tests, not a verified rubric. Searches for the exact problem number and MRPL did not establish an authoritative match; SIH pages were inaccessible. That does not prove the brief is wrong. Obtain the official statement/version and separate mandatory capabilities, illustrative examples, demo instructions and judging criteria. Until then, label R1-R6 as the team's interpretation. Absence from an example does not establish that handwriting, P&IDs or other output formats are ungraded.

“One config file makes every model interchangeable” is incomplete. You also need compatible chat templates, image preprocessing, context limits, tool-call parsing, schema support, quantization, memory scheduling and failure semantics. Keep model IDs in configuration, but expose capability contracts and run a conformance test whenever a model changes.

“A chatbot does none of these” is an unhelpful product taxonomy. Existing chat interfaces can front agents, execute tools and return files. Describe the concrete work you complete instead.

The shadow-AI claim is plausible but unverified here. The brief supplies no MRPL incident evidence showing staff actually paste confidential data into public tools. Present this as a risk or stakeholder hypothesis, not an established behavioral finding. Similarly, a local alternative can improve adoption, but its existence alone does not prevent employees using other services.

Replacement positioning: “An offline inspection workbench that links every critical reading and recommendation to visible source evidence, handles uncertainty explicitly, and produces a verified approval-note draft on a 12GB workstation.” This is a proposed product promise; earn it through the tests on pages 10-12.

Sources: Mintplex Labs, [MIT source license](#), current repository; Onyx, [Code Execution](#), undated documentation, and [release notes](#), including v4.6.0, 20 Aug 2026. Dynamic documentation is evidence of documented capabilities, not a deployment audit.

OCR: use a small, properly specified comparison

Retain Paddle as the baseline, but pin the actual version. PaddleOCR-VL-1.6 was released on 28 May 2026 and its model card lists Apache-2.0. In its paper's common OmniDocBench v1.6 comparison, the overall scores are 94.93 for Paddle-VL-1.5, 95.22 for GLM-OCR and 96.33 for Paddle-VL-1.6. These are source-reported composite parsing scores, not percentages of refinery measurements correctly extracted. The evidence does not justify announcing GLM as a better replacement for current Paddle.

Candidate	Precise role	Why it deserves a test
PaddleOCR-VL-1.6	Official layout + region-recognition pipeline	Strong current baseline; upgrade older Paddle before replacing it
GLM-OCR, 0.9B	Local layout/recognition pipeline; separate direct-schema extraction experiment	Compact challenger with structured extraction; MIT weights, Apache-2.0 code
PP-StructureV3 + pinned OCR components	Cell geometry, OCR and table reconstruction	More granular coordinates; useful for measurement provenance
Docling standard pipeline + named OCR engine + TableFormer mode	Document conversion and structured provenance	Tests full integration, including valid native PDF text

The most interesting challenger is a method, not a model. Paddle documents that PP-StructureV3 exposes finer table-cell/text coordinates than its VL series. It can reconstruct table HTML from cell geometry and rerun OCR within predicted cells. Compare that path with VL-1.6 table parsing, then test a selective cascade that rereads only disputed critical regions.

GLM's authors report stronger tables and throughput than Paddle-VL-1.5 in their own evaluations. That comparison is against an older Paddle release and is not an RTX 4070 result. It supports inclusion in the experiment, not a local speed forecast.

Do not compare raw TrOCR against a full-page parser. The handwritten TrOCR card describes single-line images. It needs segmentation before a fair page-level comparison. Docling, conversely, is a configurable conversion pipeline rather than one OCR model; naming it without its component settings makes results irreproducible.

Run two tracks: genuine scan images with no hidden text access; and mixed production PDFs where validated native text can be used. A good native-text path can avoid needless recognition errors and GPU work, but some embedded text layers are wrong and require visual checking.

Sources: PaddlePaddle, [VL-1.6 model card](#), May 2026; Zhang et al., [VL-1.6 paper, Table 2](#), 2 Jun 2026; Z.ai, [GLM-OCR repository](#), current; Duan et al., [GLM-OCR report](#), Mar 2026; PaddlePaddle, [PP-StructureV3 guide](#), current; Docling project, [conversion toolkit paper](#), Jan 2025; [TrOCR model card](#), Microsoft namespace, referencing 2021 research.

Make the hidden-thickness trap a data problem

The current flow asks the reasoning model to infer whether escalation is needed from extracted material. That invites a selective-reading failure: the prose sounds reassuring while a separate table contains the critical value. **Completeness checks should force the measurement table into the decision path.**

Use a canonical record for each reading: document hash; page; table, row and column; bounding box; equipment/CML identifier; raw OCR string; parsed decimal; unit; date; applicable minimum and its source; extraction model/configuration; and review status. Preserve the original image. A summary or a normalized Markdown table is not sufficient provenance.

Recommended flow: **page classification** → **extraction** → **row association** → **field validation** → **approved-rule evaluation** → **evidence-grounded drafting** → **artifact checks** → **human disposition**. For mixed or malformed tables, completeness checks should detect dropped rows, repeated headers, orphan values and missing units.

Use decimal arithmetic and explicit unit conversions. For a synthetic example, a verified reading of 4.8 mm and an applicable approved minimum of 5.0 mm trigger the configured below-minimum review rule. The note should show both values and their sources. These numbers are illustrative, not an engineering limit or a claim about MRPL practice.

Return three statuses: **rule-triggered escalation**; **no trigger under the evaluated rules**; **unresolved evidence requiring review**. Missing or ambiguous measurements must not become “safe.” Nor does being above one threshold establish fitness for service. The real rule set needs the asset, relevant standard edition, operating context and authorized engineering interpretation. Do not let the model invent a minimum thickness from general knowledge.

Selective rereading: start with the chosen extraction pipeline; if a critical field is missing, two readings disagree, row linkage is uncertain, or the result is near a configured boundary, reread the original cell/table crop through a second path. Retain alternatives. If unresolved, show the crop to the reviewer. Do not silently repair a digit because a more plausible value makes the report consistent.

OCR confidence, model self-confidence and agreement between two recognizers are signals, not correctness probabilities. Calibrate acceptance on held-out target-like data and report the error rate among accepted fields alongside the fraction sent for review. Disagreement routing may save latency relative to running two parsers on every page; it can also miss shared errors. Both outcomes belong in the comparison.

The resulting advantage is inspectable: click the recommendation, see its calculation and applicable rule, then see the exact source cell. The language model remains useful for terminology and drafting, while the critical path is reproducible.

Sources: PaddlePaddle, [PP-StructureV3 capabilities](#), current README; Docling project, [document model and provenance](#), current. The schema, workflow, example and acceptance rules above are this review's engineering recommendations, not benchmarked outcomes or an interpretation of a licensed inspection code.

A realistic model plan for the RTX 4070

Test one useful general model before building a model fleet. Qwen3.5-9B is an Apache-2.0 multimodal candidate for extraction follow-up, tool use, explanations and drafting. Qwen3.5-4B is a smaller fallback candidate. These are bounded choices, not a claim that either is the best model available. Official model-card benchmarks are not evidence of reliable MRPL tool execution after quantization.

Use Qwen2.5-Coder-7B-Instruct as a coding-specialist control, not an automatic winner because it says “Coder.” Compare it with the general candidate on your actual sandbox tasks using the same tests, repair budget and deployment constraints. Keep it only if the extra model improves verified task completion enough to justify loading and integration cost.

Role	Initial candidate	Deployment hypothesis
General + image follow-up	Qwen3.5-9B, then 4B if needed	Tested 4-bit artifact; short controlled context first
OCR	PaddleOCR-VL-1.6 or winning challenger	Separate local worker; release GPU allocations after its stage
Coding control	Qwen2.5-Coder-7B-Instruct	On-demand loading; compare against general model
Dense retrieval	Qwen3-Embedding-0.6B	CPU-first experiment; precompute corpus embeddings
Optional reranking	Qwen3-Reranker-0.6B	Only a small candidate set; benchmark CPU latency

Weight arithmetic is a lower bound, not a VRAM estimate. At exactly 4 bits per parameter, 9B weights occupy about 4.5 GB and 7B about 3.5 GB, using decimal GB. Scales, higher-precision tensors, vision components, runtime buffers, activations, cache and display usage add memory. “Both files fit” does not establish that both running models fit. A 30B MoE at the same nominal precision needs about 15 GB for weights alone; active parameters reduce compute, not the total stored weights.

This matters for Sarvam: the 30B card lists Apache-2.0 and Indian-language focus, with 2.4B active non-embedding parameters. That does not make it a 2.4B-memory model or establish a comfortable all-GPU 12GB deployment. CPU offload or a smaller translation model can be tested if the corpus warrants it. Language, translation, OCR and reasoning requirements should be measured separately.

Use explicit GPU ownership and stage batching: OCR pages, release OCR, draft, then load a coding specialist if required. Test warm/cold loading, 8K versus 16K application context limits, peak total VRAM and p95 end-to-end latency. Those limits are experiment settings, not the models' native maximum or promised quality levels.

Sources: Qwen official model cards, current, [Qwen3.5-9B](#), [4B](#), [Coder-7B](#), [Embedding-0.6B](#), [Reranker-0.6B](#); Sarvam AI, [Sarvam-30B](#), current. Memory figures are arithmetic estimates, not measurements.

Simplify orchestration; strengthen the contracts

The inspection demo is mostly a known workflow. A general Thinker delegating to multiple Workers and a Verifier adds opportunities for omitted evidence, repeated model calls and repeated side effects. Start with explicit states and one bounded tool-using agent where adaptive choices actually help, such as choosing a reread or refining a retrieval query. Keep a more autonomous loop for the coding task if useful.

This preserves meaningful agent behavior. The model can select tools and recover from tool errors inside a controlled boundary; the application owns permitted transitions, resource limits and artifact publication. If the official statement requires model-driven planning specifically, demonstrate that bounded planning rather than disguising a fixed script as an autonomous agent.

Compare a small Python state machine plus typed validation with LangGraph only after you have the same working vertical slice. LangGraph's persistence can help with interruptions and recovery; its in-memory checkpointer does not survive a process restart. A framework does not automatically make writes idempotent. Persist input hashes, outputs and state transitions; on retry, resume or repair the failed stage instead of creating duplicate files or approvals.

Pydantic AI is a reasonable typed-agent candidate; smolagents remains an alternative if its execution approach fits your tools. There is no evidence in the brief that running a four-framework tournament now will improve the demo. Shortlist by one injected failure: interrupt after extraction, restart offline, and finish without repeating successful work or losing source links.

Router improvement: use workflow stage, required capabilities, resource availability and previous failures. “An image exists” should trigger document inspection, not force every subsequent call to a vision model. “Running code” does not require an LLM at all; generating or repairing code may. Record which rule fired and the chosen model. If R2 demands two task-specific LLMs, confirm whether OCR plus general reasoning counts before relying on that interpretation.

Serving recommendation: try llama.cpp first for the small quantized general/coding models and use the OCR project's supported local runtime separately. Its server documents model loading/unloading and schema-constrained output. Runtime routing dispatches a selected model; it does not understand the user's task. Ollama is a sensible usability control. Move to vLLM/SGLang when compatibility, concurrent throughput or a measured bottleneck warrants the extra comparison.

Pin model artifact hash, runtime version, tokenizer/chat template, preprocessing, precision, context, tool parser and decoding settings. Test image input, schema output, tool calls, timeouts, unload/reload and a cold offline restart. Schema validity checks format; they do not verify the truth of a measurement.

Sources: Anthropic, [Building effective agents](#), 19 Dec 2024; LangChain, [LangGraph persistence](#), current; ggml-org, [llama.cpp server documentation](#), current. Architecture preference is an inference for this workload, not a measured framework ranking.

Retrieval: exact clauses before a knowledge graph

The proposed comparison omits an essential baseline: **lexical retrieval**. Equipment tags, clause numbers, document revisions and abbreviations often require exact matching. Dense similarity alone may find the right subject while missing the exact applicable clause. Two-document questions do not automatically require a knowledge graph.

Start with clause-aware chunks carrying document ID, title, edition/revision, effective date, section path, page and permissions. Preserve table headers with rows and expand to the parent clause when necessary. Search BM25/lexical and dense indexes in parallel, merge with reciprocal rank fusion, and optionally rerank a bounded candidate set. Qdrant documents this pattern; it is an implementation option rather than a requirement to add another server.

Test BM25-only, vector-only, their fusion, and fusion plus reranker on the same labeled questions. An exact clause/tag lookup can bypass broad semantic search. For cross-document questions, try explicit query decomposition and follow visible references before adding inferred graph edges.

Microsoft GraphRAG supports global corpus summaries and local entity-centered retrieval. Those are legitimate uses. They do not establish that graph construction will improve this project's clause lookup. A graph adds extraction/index maintenance and creates another place for unsupported relationships to enter the evidence chain. Keep it if a held-out relationship-heavy slice gains enough evidence recall and answer quality to offset those costs.

Citation resolution is only the first check. Confirm that the source exists, the cited span supports the claim, the edition applies, and the asset/context matches. A correct URL to the wrong clause is still a wrong citation. Distinguish an observation from an inspection report, a normative rule from an approved standard, and a recommendation inferred by the assistant. "Every claim cites a clause" is too narrow for measurements and too permissive for invented reasoning.

In the draft, make critical conclusions a structured combination of approved evidence records and rule outputs. Limit the model to retrieved citation IDs rather than letting it invent clause numbers. For free text, check citation support and allow "insufficient evidence." An entailment model may help triage, but is not the final authority on engineering applicability.

Permissions are a data-model requirement now. A single-user, curated demo can defer enterprise identity integration. Still store permissions and apply filters before content enters retrieval, generation and citation previews. Do not mix restricted vendor documents into a supposedly safe shared corpus and promise to add access control later. In production, include derived indexes, caches, logs, previews and backups in the same policy.

Sources: Qdrant, [Hybrid and multi-stage queries](#), current; Microsoft, [GraphRAG query overview](#), current; Gao, Yen, Yu and Chen, [Enabling LLMs to generate text with citations](#), EMNLP, Dec 2023. Proposed corpus-specific benefits and permission design remain engineering recommendations.

Verification should inspect truth, structure and presentation

Keep rendering; remove the idea that a vision-model pass certifies the file. A VLM may notice overlap or a broken table, but cannot prove that every number, formula and citation is correct. Running another prompt is not an independent oracle, especially when it sees the same mistaken extraction.

Use four checks with different evidence:

Check	What to verify	Useful failure response
Input/decision integrity	Critical fields match accepted evidence; all required rows covered; rules recompute	Reopen the disputed extraction or rule input
Artifact structure	File parses; required sections/rows present; no placeholders; links and field values preserved	Repair the template/render stage
Semantics	Claims have support; units, dates, assets and applicable rules align	Remove or mark unsupported text; seek review
Presentation	Rendered pages have no clipped content, broken pagination or unreadable tables	Fix layout and render again

Use python-docx with an approved template for the initial note. Populate repeated factual fields and measurement tables directly from the canonical record. Keep model-generated prose in constrained sections. Read the resulting DOCX back and compare critical values with the record. Inspect both structure and rendered output; opening successfully is a very weak acceptance test.

For spreadsheets, openpyxl writes formulas but does not evaluate them. If XLSX is added, use a pinned compatible calculation engine such as a tested local office suite, then compare important computed results with independent reference calculations. Recalculation alone can faithfully compute the wrong formula. Word does not need spreadsheet recalculation; apply checks appropriate to each artifact type.

Route failures to their cause. A clipped footer should not restart engineering reasoning; a disputed thickness digit should not be “fixed” by rewriting the approval note. Bound retries by stage, preserve the failed output and return a reviewable incomplete result when the budget is exhausted.

The coding demo needs known expected behavior. Choose a small task such as parsing a CSV of verified readings, handling missing/invalid rows, computing a configured comparison and writing a summary. Test normal input, boundary equality, missing values, wrong units and a malformed row. Include at least one held-out test the model did not write. An agent that produces its own code and its own passing tests has not established correctness.

Your output should be an **approval-note draft with a verification record**, not an autonomous engineering approval. Show fields needing review and the checks actually performed. This makes the workflow more useful without claiming that a layout check or a local model authorizes an operational decision.

Sources: openpyxl project, [Simple Formulae documentation](#), current 3.1.3 documentation; Gao et al., [citation evaluation research](#), Dec 2023. Layered checks, test cases and repair routing are recommendations developed for the brief.

Security: defend the actual boundary

Reject the assumption that code from your model is trusted. An offline document can contain instructions that redirect a tool-using agent. A trusted user can upload an untrusted vendor file. Without network access, a bad program can still corrupt inputs, alter drafts, read unrelated files or exhaust resources. Indirect prompt injection is an established attack class; this does not mean the specific prototype has been attacked.

Docker can remain the prototype baseline, but justify it through explicit controls: fresh task container; non-root/rootless where supported; no network; dropped capabilities; no-new-privileges; seccomp and an available host MAC policy; read-only inputs/root; narrow writable outputs; no host home or Docker socket; memory, CPU, PID, time and output-size limits. Preload dependencies. Keep the GPU model service outside this CPU tool sandbox. Container controls reduce risk but do not establish complete kernel isolation.

Test gVisor as a stronger CPU sandbox if required or easily compatible. It reduces the host-kernel attack surface, but still requires file, network and resource restrictions. MicroVMs are an option, not automatically mandatory for this bounded demo. Prefer an approach you can configure and test over an isolation label.

A firewalled workstation is not automatically physically air-gapped. For the demo, native Linux simplifies the boundary. WSL can work, but a Linux packet capture alone does not cover the Windows host. Microsoft documents DNS tunneling through virtualization rather than ordinary Linux network packets; include Windows, WSL/Hyper-V, IPv4/IPv6 and DNS/proxy paths in the stated observation scope.

Separate approved local traffic, external attempts, denied attempts and observed successful external egress. Your deliberate canary creates an attempted external call; it cannot coexist with a literal “no external call ever made” claim. Use a separately marked canary window, then clean startup and task windows. A packet dropped before the capture point may appear only in firewall or connection-attempt logs.

The precise Ollama claim needs its tested version and logs. Current desktop updater source targets Windows/macOS and defines a 3-second initial delay, hourly interval and an ollama.com update URL. That does not substantiate the universal claim that every Ollama server contacts GitHub after four seconds. AnythingLLM also documents telemetry controls; audit the whole pinned build, including the UI.

Demonstrable claim: “This build completed these tasks with external egress blocked; the stated monitors observed no successful external egress during the recorded run.” Save the configuration, process/interface inventory, build hashes, logs and capture. Include a cold offline start, fonts, model/tokenizer caches, document renderers and tool dependencies. Treat evidence logs as confidential data too.

Sources: Zhu et al., [MELON / indirect prompt injection](#), ICML 2025; Docker, [Engine security](#); gVisor, [security model](#); Microsoft, [WSL networking](#); Ollama, [updater source](#); AnythingLLM, [privacy and data handling](#). Software sources are current as accessed 6 Sep 2026.

Your evaluation needs independent cases and better targets

The existing corpus is a useful debugging seed, not validation of deployment reliability. Twelve reports transformed four ways remain twelve underlying reports. Keep every version of a report in the same split. Hold out document templates, authors and acquisition conditions too; otherwise a system can succeed by recognizing the generator's layout.

Do not describe the real PDFs as self-grading without qualification. Embedded text can have broken order, missing characters or incorrect table associations. Rendering and comparing text measures recognition against that text layer; it does not supply verified row/cell labels or the correct escalation decision. The CSB report is useful contextual material, but it is not a completed MRPL inspection form or the operative standard. The unidentified tender remains unverified here.

Create three sets: development cases for tuning; a frozen template-held-out test set; and a separately identified challenge set. Add real scans or photographs with independently checked critical fields. Ask a domain reviewer to label the expected disposition and applicable evidence where possible. Generate counterfactual pairs that change only one digit, unit, equipment ID or threshold source. Balance positives for diagnosis, but also evaluate a realistic prevalence mix for operational workload.

Measure	What counts as success
Critical-field accuracy	Exact value AND unit AND correct asset/row; count missing fields explicitly
Escalation performance	Report false negatives, false positives and unresolved/review cases separately
Selective accuracy	Error rate among accepted fields AND percentage requiring review
Retrieval	Relevant-clause recall at k, correct edition, evidence completeness and permission filtering
Deliverable quality	First-pass correct files, critical-value preservation and time until accepted output
Efficiency	p50/p95 total latency, cold load, peak VRAM/RAM, retries and review minutes

CER and table-structure scores remain diagnostic metrics. They should not select a model that reads boilerplate perfectly but swaps two thickness rows. A system that sends every case to a human must not win merely by making no automatic errors. Measure review workload and useful completion coverage.

Assuming the reported balanced corpus has six independent escalation-positive reports, zero misses still gives an exact one-sided 95% binomial upper bound of about **39.3%** on the miss probability: $1 - 0.05^{(1/6)}$. Even zero misses in 300 independent representative positive cases gives about **0.994%**, not zero risk. These calculated examples illustrate sample uncertainty; correlated synthetic reports do not satisfy the independence/representativeness assumptions.

Freeze prompts/configuration before the test; use the same pages, resolution, tool contracts and retry budget across candidates. Also allow each model a separately reported, equally budgeted development tuning pass. Record quantized-artifact hashes. Report paired results and uncertainty at the source-document level; repeated runs measure stochastic stability, not extra independent documents.

Sources: US Chemical Safety Board, [Chevron Richmond refinery investigation](#), final report released 28 Jan 2015. Corpus counts are from the supplied brief and were not audited. Evaluation design and statistical calculations above are this review's analysis.

Handwriting, drawings and language: narrow the promise

“Handwriting is unsolved” is too absolute. The closest primary match for the brief’s error claim is METATR, which reports 28.5% handwritten CER for Gemini 3 Pro. Its 29-language collection includes historical scripts, difficult layouts and degraded material. The brief’s original citation was not supplied, so this is a likely related source, not confirmed provenance. It does not establish MRPL handwriting accuracy or a universal performance ceiling. CER includes insertions and deletions and is not simply the fraction of characters misclassified.

A defensible position is: unrestricted handwriting is outside the committed demo pending representative evidence. Test handwritten dates, tags and numeric fields in known regions if those matter operationally. Compare line/cell recognition and human-assisted correction. Domain adaptation might help after sufficient labels exist; twelve synthetic printed reports cannot establish that it will. Do not claim calibrated confidence until it is measured on target-like handwriting.

Keep complete P&ID reconstruction as stretch, but separate it from useful drawing support. Microsoft’s reference implementation exists and depends on Azure services. That supports the “local rework required” statement, not a universal assertion that no tools exist. Symbol recognition, equipment-tag localization, line connectivity and engineering interpretation are different problems. A useful smaller feature is retrieving the right drawing revision, locating a tag and showing its source region for human confirmation. Do not market that as complete topology reconstruction.

Language choice should follow representative documents and user tasks. English prose with Kannada annotations, Hindi correspondence and equipment codes is not one generic “multilingual” requirement. Check recognition, retrieval and drafting separately, and preserve original-language evidence when translating. Sarvam’s local weights and Indic focus justify a candidate evaluation; they do not make a cloud API acceptable inside the stated offline boundary or guarantee local fit.

Do not build every Office format just because the libraries exist. Word is the best initial complete artifact if the official requirement confirms it. If PowerPoint or Excel is mandatory, add one narrowly templated example with its own verification rules. There is little value in broad file-format coverage that repeatedly needs manual repair.

Reuse an application, but evaluate it as a competitor and integration dependency. AnythingLLM is a strong first trial because its router and file generator already overlap your plan. LibreChat is a credible alternative: current documentation supports a self-hosted code interpreter. Neither product’s documentation proves your pinned configuration works offline or passes the measurement workflow.

Run a one-day integration trial: local endpoint, scan upload, tool-returned findings JSON, downloadable DOCX, failed-tool recovery and audit events. Keep the option requiring less adapter code and preserving your evidence links. If neither fits, a small upload/status/download interface may cost less than bending a full chat product to the workflow.

Sources: Boillet, Tarride and Kermorvant, [METATR](#), 26 May 2026; Microsoft Azure Samples, [P&ID digitization repository](#), current; LibreChat, [self-hosted code interpreter documentation](#), current. Suggested narrower product capabilities are hypotheses to validate with MRPL users.

Build the evidence before expanding the stack

Change the order of work now. The brief reports only research and a small generator as completed. Before another broad technology comparison, finish one thin path using the current best baseline. Then replace one component at a time.

Stage	Concrete deliverable	Advance when
1: contract and baseline	Official requirement copy; canonical field schema; fixed sample; Paddle → rule → templated DOCX	Critical values round-trip; missing evidence produces review status
2: complete vertical slice	Evidence crops, retrieval, render checks, run manifest, offline dependencies	Cold start and full task complete offline; failure is visible and resumable
3: extraction comparison	VL-1.6 vs GLM vs cell-based path; frozen real/held-out cases	Winner selected by critical errors, coverage and review time
4: bounded intelligence	General-model comparison, coding control, stage router and repair budgets	Routing logged; coding passes independent tests; memory stable
5: deployment evidence	Hardened sandbox, canaries, network boundary, UI integration	Prohibited actions fail; legitimate task succeeds; logs explain both

Recommended first experiments: **A**, compare LLM-only escalation with the same extracted inputs fed to explicit approved rules; **B**, compare full-table parsing with cell-based extraction and selective rereading; **C**, compare vector-only with BM25 + vector and optional reranking; **D**, compare the general model with the coding specialist under an identical repair budget. These isolate causes better than changing the model, framework and OCR pipeline simultaneously.

Set replacement criteria before running the held-out set. Prefer fewer critical misses at comparable review burden. If quality is tied within the available evidence, choose lower end-to-end latency, smaller memory demand and less maintenance. A proposed screening target is a 20% latency or review-time reduction with no observed increase in critical errors; this is a team decision threshold, not an established significance test or promise. If confidence intervals are wide, report that the winner is provisional.

The live demo should show: the hidden low-thickness case; a genuinely ambiguous field routed for review; the cited source and rule; a correct Word draft; one failed coding attempt repaired against a test; and the separately marked network canary. Show observed actions and results rather than a decorative stream of agent thoughts. A source-crop panel is a worthwhile small custom interface even when chat is reused.

Defer full P&ID topology, broad handwriting, fine-tuning, a large inferred knowledge graph, simultaneous resident specialist models and a four-framework tournament. Reopen each only when a measured error family or confirmed requirement justifies it. Retain a locked known-good build before the demonstration.

What remains unknown: official grading wording; actual corpus languages and handwriting; venue GPU/host RAM/OS; access policy; licensed standards and their applicable editions; actual model versions and packet logs; and performance on real MRPL documents. None can be settled by another public leaderboard. This review stops at a ranked, testable architecture decision: the next evidence must come from your corpus and workstation.

Research record: primary-source discovery was followed by targeted contradiction checks for competitors, OCR versions, handwriting scope, serving contracts, security and verification. High-impact claims were reread in the coordinating review. No production deployment, legal license opinion, engineering certification or local performance result is asserted.